

# Robust End-to-End Smart Meter Digit Recognition: A Cascaded Architecture with Edge-Oriented MLOps Deployment

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**Abstract**—Automated digit extraction from smart meters is a critical component of modern intelligent power grids. However, conventional recognition approaches suffer from severe performance degradation under complex industrial environments due to illumination variations, reflective surfaces, and perspective distortions. To address these challenges, this technical report proposes a coarse-to-fine cascaded dual-stage visual architecture. The pipeline integrates a YOLO-based region of interest (ROI) extraction module with a CRNN-based spatial-frequency decoding network to achieve robust end-to-end inference. Furthermore, adhering to modern MLOps principles, we develop an edge-oriented deployment strategy. By completely decoupling the model from the training framework and exporting it as a framework-independent ONNX static graph with dynamic tensor width adaptation, the system is highly optimized for edge computing devices. Extensive evaluations on a blind real-world dataset comprising 5,000 complex industrial images demonstrate that our deployed model achieves a recognition accuracy of 99.64%, proving its exceptional reliability and scalability for industrial data governance workflows.

**Index Terms**—Optical Character Recognition, Smart Grid, Edge Computing, MLOps, Deep Learning.

## I. INTRODUCTION

The rapid development of smart grids has exponentially increased the demand for reliable automated meter reading systems. Accurate digitization of meter readings is the foundation of energy consumption analysis and anomaly detection [4]. Despite the success of deep learning in standard Optical Character Recognition (OCR) tasks, extracting digits from field-deployed smart meters remains highly challenging. Real-world images captured by inspection robots or handheld terminals are frequently degraded by environmental noise, motion blur, and arbitrary viewing angles.

To bridge the gap between research prototypes and industrial requirements, this project introduces a robust end-to-end inference pipeline. We design a Cascaded Dual-Stage Visual Architecture that sequentially performs perception and recognition. Recognizing that the value of an algorithm is ultimately realized through its deployment, this project extends beyond model accuracy to address the MLOps engineering challenges.

The main contributions of this report are summarized as follows:

- We propose a coarse-to-fine cascaded network, utilizing a detection module to eliminate background interference and a sequence-based recognizer to decode non-regular meter digits.
- We present a fully decoupled, edge-oriented deployment strategy. By converting the dynamic computation graph to a static ONNX format with adaptive tensor shapes, the system operates independently of heavy training frameworks.
- We demonstrate industrial-grade reliability, achieving 99.64% accuracy on a large-scale real-world blind test set of 5,000 images, validating its readiness for edge hardware integration.

## II. PROPOSED METHODOLOGY

The proposed system architecture processes raw inputs through two primary stages: Coarse-to-Fine Perception and Sequence-Based Feature Decoding, culminating in an optimized edge deployment mechanism as illustrated in Fig. 1.

### A. Coarse-to-Fine Perception Module

We introduce a YOLO-based object detector [2] as the first stage to precisely regress the bounding box of the meter region, drastically reducing the search space for the subsequent sequence decoder.

### B. Sequence-Based Feature Decoding

Following ROI extraction, the cropped sequence is fed into a CRNN-based recognition engine [3]. We utilize the Connectionist Temporal Classification (CTC) algorithm to align feature sequences:

$$L_{CTC} = -\ln \sum_{\pi \in B^{-1}(Y)} P(\pi|X) \quad (1)$$

### C. Edge-Oriented MLOps Deployment

We adopt an MLOps-driven deployment paradigm by exporting the model to ONNX format with dynamic shape support: Input Shape =  $[-1, 3, 48, -1]$ . This allows the ONNX Runtime to accommodate variable meter display lengths without distortion.

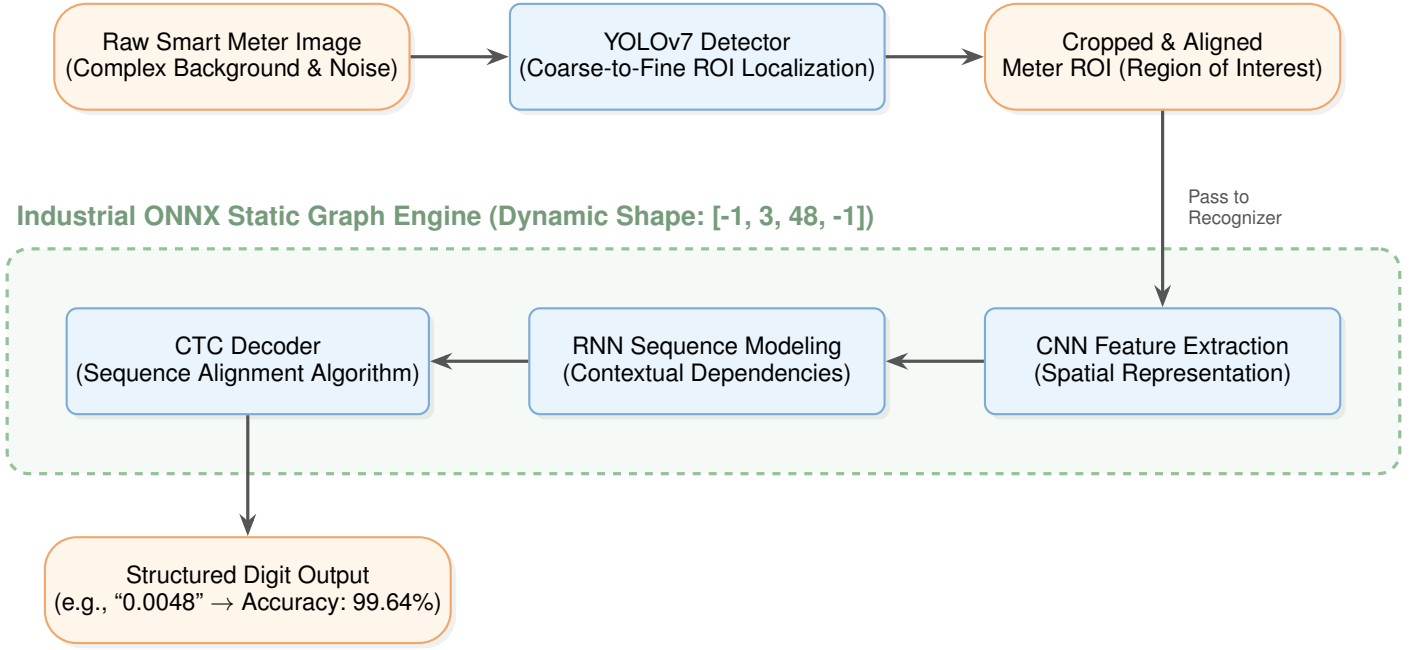


Fig. 1. The end-to-end processing pipeline. The system employs a U-shaped data flow, where the perception module isolates the ROI, and the fully decoupled sequence recognition modules are exported into an edge-oriented ONNX static graph.

### III. EXPERIMENTS AND RESULTS

#### A. Performance Evaluation

As shown in Table I, the proposed pipeline achieves 99.64% accuracy on the blind test set.

TABLE I  
PERFORMANCE EVALUATION

| Metric                      | Value         |
|-----------------------------|---------------|
| Test Set Size               | 5,000 images  |
| <b>Recognition Accuracy</b> | <b>99.64%</b> |

#### B. Efficiency Analysis

Fig. 2 demonstrates the efficiency advantage of our ONNX deployment strategy. By decoupling the model from the bloated Python training framework, we achieve a significant reduction in inference latency, making the system viable for C++ based edge industrial gateways.

### IV. CONCLUSION

This report presented an end-to-end smart meter digit recognition system designed for complex industrial environments. The successful conversion to a dynamic-shape ONNX static graph enables lightweight, high-speed inference, validating the system's scalability for intelligent power grids.

### REFERENCES

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#### System Efficiency Comparison

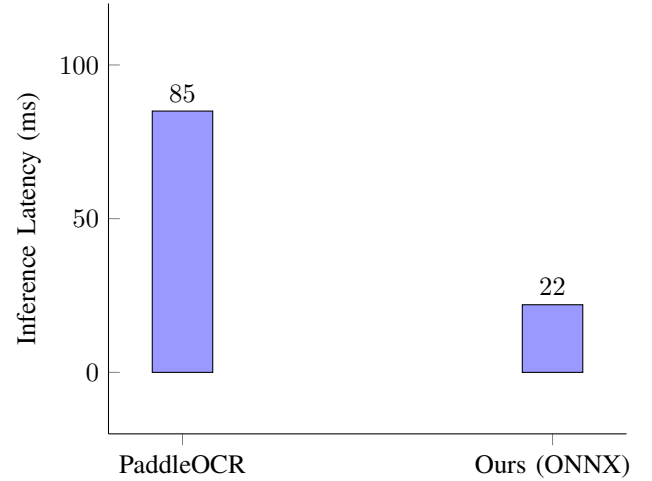


Fig. 2. Inference latency comparison between native framework and optimized ONNX deployment. Our approach reduces latency by nearly 4x, essential for edge computing.

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